

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	40	320
human-ai	100	751
human-human	36	288

3 Preprocessing

Slot-level rows: 2618

Metric rows: 7854

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from $\text{resonance} \sim \text{surprise} *$ slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	0.928	0.868	-0.060	0.041	-1.479	0.139	0.060	
Human-Human	0.909	0.885	-0.024	0.059	-0.405	0.686	0.024	
AI-AI	0.981	0.886	-0.096	0.053	-1.814	0.070	0.096	.

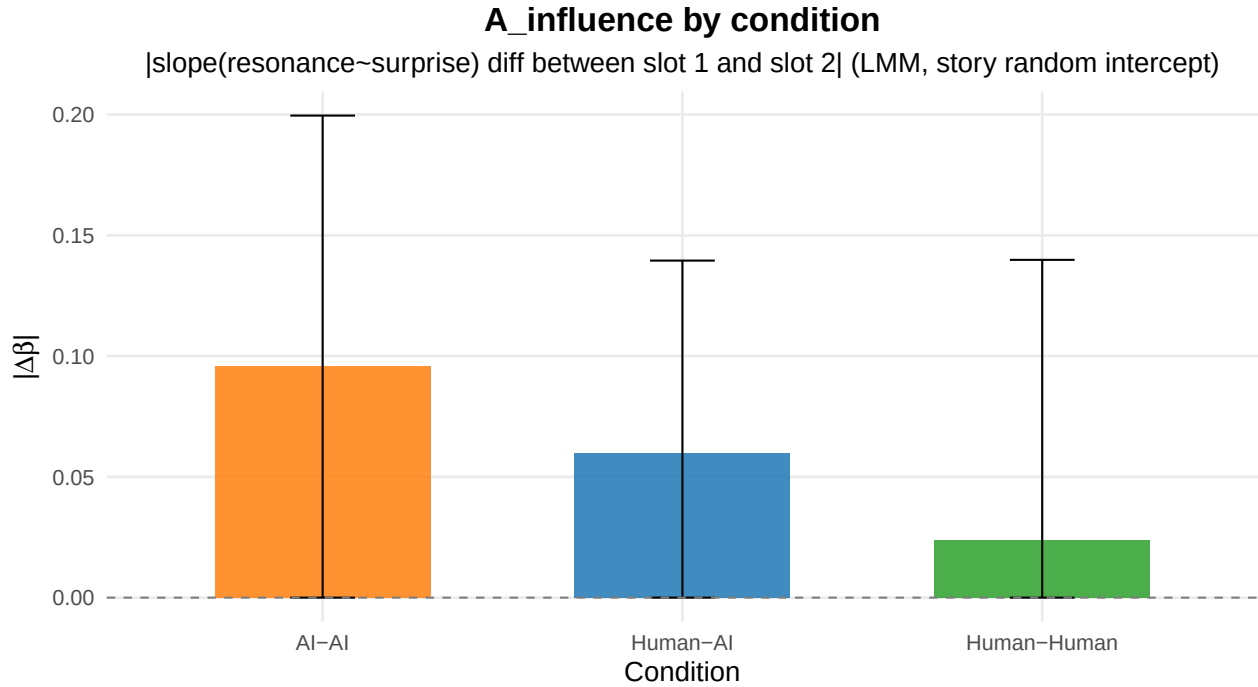


Table 3: Pooled model: |slope_slot1 - slope_slot2| (unsigned) by condition

condition	A_influence	se
Human-AI	0.069	0.041
Human-Human	0.023	0.060
AI-AI	0.100	0.054

Table 4: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	100	0.402	0.428
Human-Human	36	0.446	0.474
AI-AI	40	0.335	0.400

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 0.54598, df = 2, p-value = 0.7611
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
```

```
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##           Human-AI Human-Human
## Human-Human 0.65      -
## AI-AI       0.65      0.65
##
## P value adjustment method: BH
```

5 Bootstrap CIs and Effect Sizes on A_influence_story

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	0.428	0.366	0.491
Human-Human	36	0.474	0.359	0.602
AI-AI	40	0.400	0.315	0.494

Table 6: Pairwise Cliff’s delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	-0.051	negligible
Human-AI vs AI-AI	100	40	0.050	negligible
Human-Human vs AI-AI	36	40	0.094	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	-0.009	0.051	-0.179	139.587	0.858	-0.11	0.092

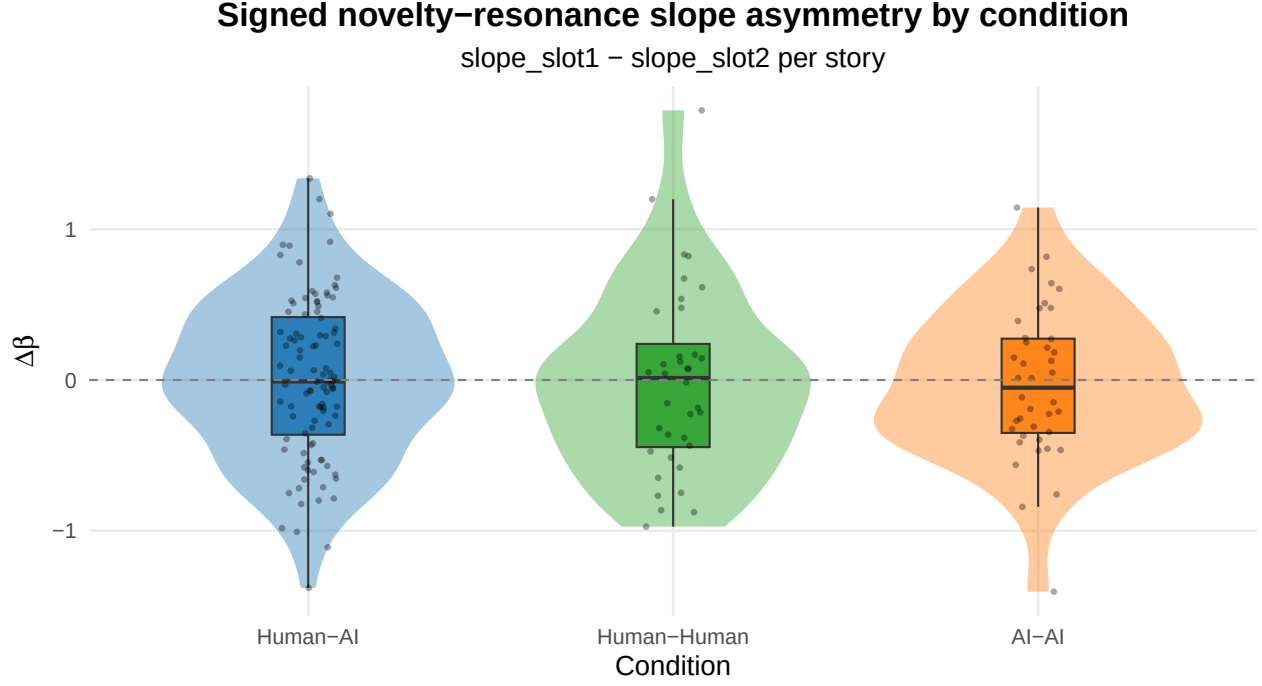
7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned A_influence is similar.

Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	100	0.011	0.536	0.020	0.51	0.838
Human-Human	36	-0.011	0.615	0.019	0.50	0.912
AI-AI	40	-0.027	0.501	0.054	0.50	0.736

Brown-Forsythe variance-equality p = 0.6079



8 Length-Controlled Variant of A_influence

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

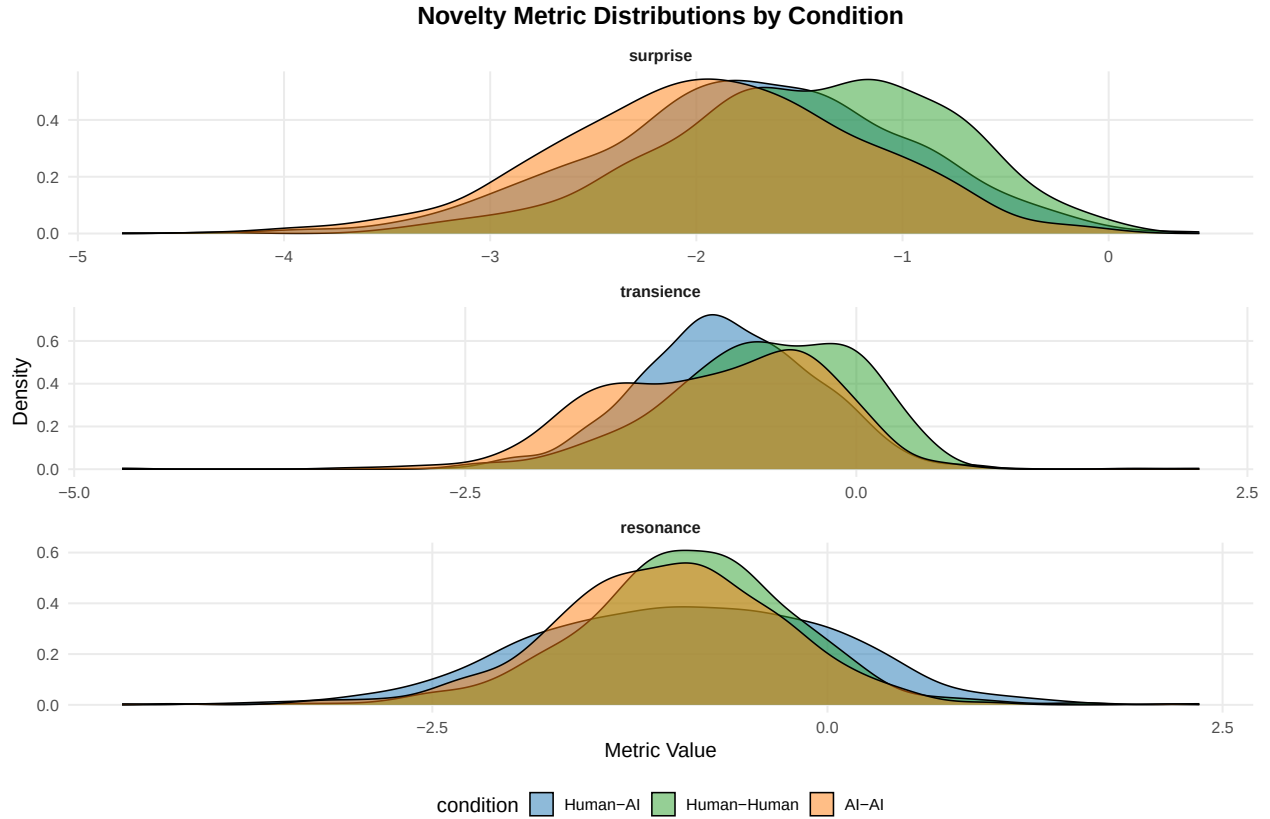
Table 9: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance $\sim \text{surprise} * \text{slot} + \text{nwords}$)

condition	A_influence_lc	se	p
Human-Human	0.019	0.059	0.747

9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	100	1402	-1.740	0.745
Human-AI	transience	100	1402	-0.833	0.563
Human-AI	resonance	100	1402	-0.907	0.929
Human-Human	surprise	36	576	-1.465	0.688
Human-Human	transience	36	576	-0.589	0.628
Human-Human	resonance	36	576	-0.875	0.711
AI-AI	surprise	40	640	-1.925	0.722
AI-AI	transience	40	640	-0.905	0.699
AI-AI	resonance	40	640	-1.020	0.735



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 5465
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8534 -0.6126  0.0158  0.6608  3.3529
##
## Random effects:
```

```

## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1036  0.3218
## Residual          0.4245  0.6515
## Number of obs: 2618, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -1.74158    0.03660 179.54384 -47.581 < 2e-16 ***
## conditionHuman-Human  0.27691    0.07038 172.06387  3.934 0.000121 ***
## conditionAI-AI     -0.18349    0.06777 172.27558 -2.708 0.007459 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.520
## condtnAI-AI -0.540  0.281
## contrast          estimate      SE  df t.ratio p.value
## (Human-AI) - (Human-Human)  -0.277 0.0704 170  -3.934  0.0004
## (Human-AI) - (AI-AI)         0.183 0.0678 170   2.708  0.0224
## (Human-Human) - (AI-AI)      0.460 0.0829 167   5.556 <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4199
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.9570 -0.5868  0.0449  0.6368  5.1353
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1216  0.3486
## Residual          0.2519  0.5019
## Number of obs: 2618, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -0.83383    0.03736 177.29830 -22.316 < 2e-16 ***
## conditionHuman-Human  0.24449    0.07218 173.03506  3.387 0.000874 ***
## conditionAI-AI     -0.07128    0.06949 173.15621 -1.026 0.306437
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.518

```

```

## condtnAI-AI -0.538 0.278
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) -0.2445 0.0722 171 -3.387 0.0026
## (Human-AI) - (AI-AI) 0.0713 0.0695 171 1.026 0.9194
## (Human-Human) - (AI-AI) 0.3158 0.0851 170 3.709 0.0008
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6528.5
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.0997 -0.6473 0.0269 0.6690 3.8439
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.0000 0.0000
## Residual 0.7054 0.8399
## Number of obs: 2618, groups: conversation_id, 176
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -0.90656 0.02243 2615.00000 -40.416 < 2e-16 ***
## conditionHuman-Human 0.03122 0.04157 2615.00000 0.751 0.45265
## conditionAI-AI -0.11341 0.04007 2615.00000 -2.830 0.00468 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.540
## condtnAI-AI -0.560 0.302
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0312 0.0416 161 -0.751 1.0000
## (Human-AI) - (AI-AI) 0.1134 0.0401 162 2.830 0.0157
## (Human-Human) - (AI-AI) 0.1446 0.0482 150 2.998 0.0095
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

```

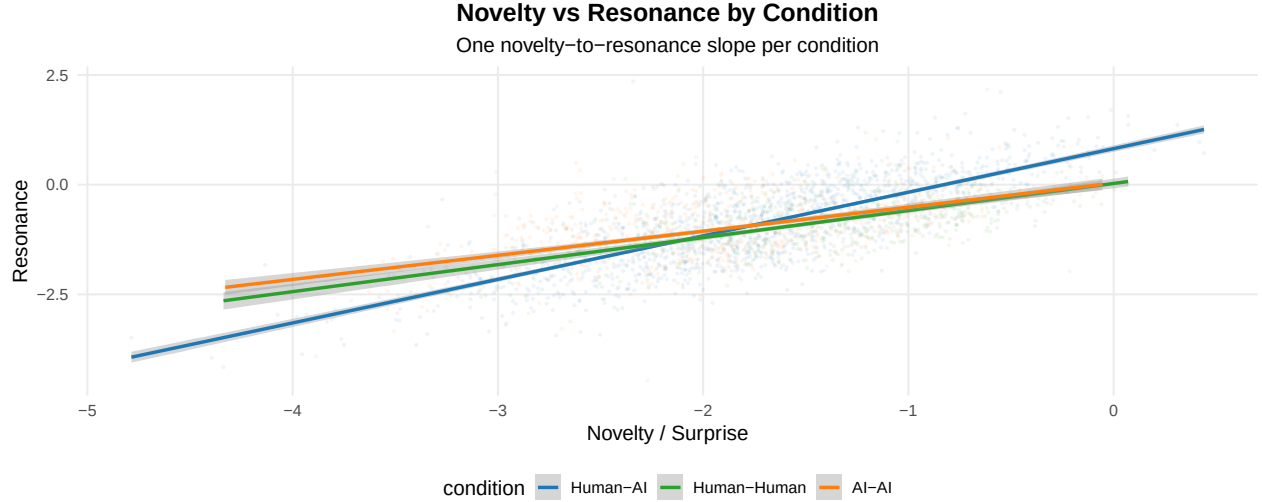
10 Supplemental: Pooled Novelty-to-Resonance Relationship

```
## Novelty-to-resonance model:
```

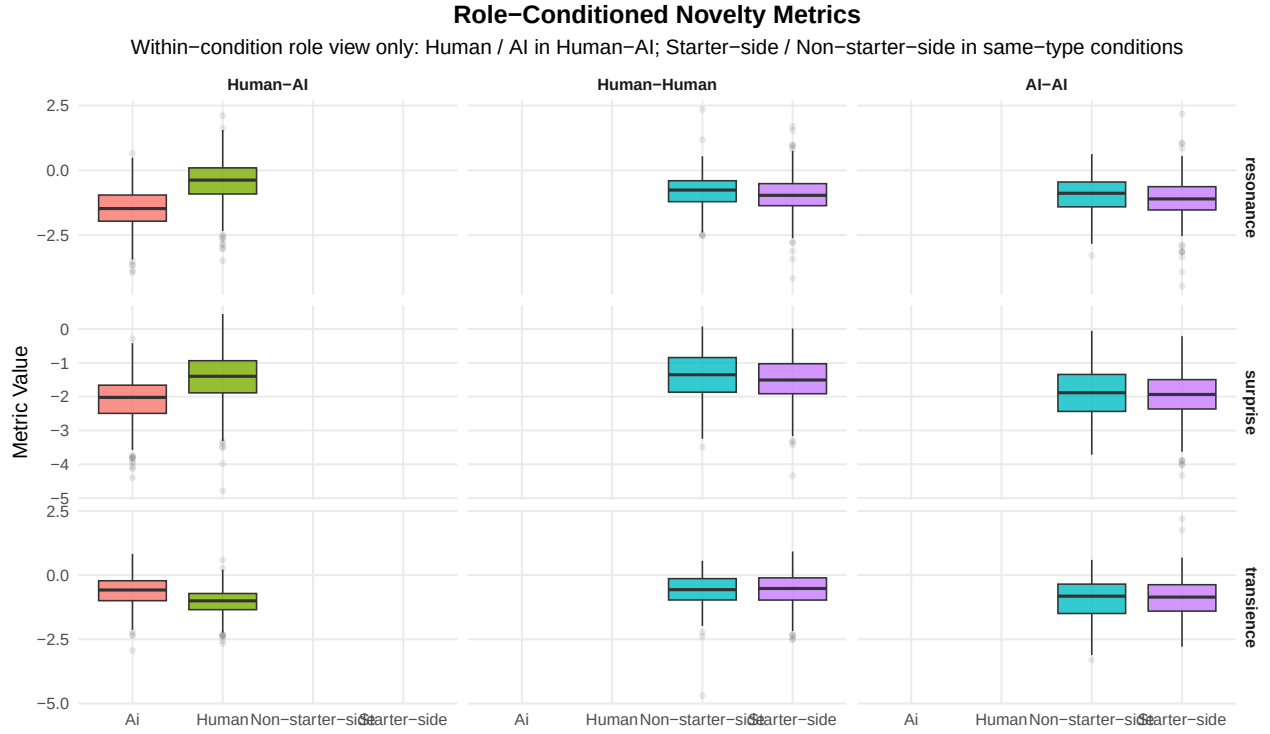
```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
## Data: df_slots
##
## REML criterion at convergence: 4183.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.2606 -0.6278 -0.0416  0.6034  6.8856
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1068   0.3268
## Residual                0.2510   0.5010
## Number of obs: 2618, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.95215    0.04860  530.52185  19.590 < 2e-16
## surprise          1.06794    0.01917 2527.90893  55.716 < 2e-16
## conditionHuman-Human -0.53251    0.09303  478.17710  -5.724 1.84e-08
## conditionAI-AI      -0.25622    0.09889  637.98150  -2.591 0.00979
## surprise:conditionHuman-Human -0.18380    0.04141 2609.57231  -4.438 9.46e-06
## surprise:conditionAI-AI -0.17660    0.03928 2611.72238  -4.496 7.22e-06
##
## (Intercept)          ***
## surprise              ***
## conditionHuman-Human ***
## conditionAI-AI        **
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.687
## cndtnHmn-Hm -0.522 -0.359
## condtnAI-AI -0.491 -0.338 0.257
## srprs:cnH-H -0.318 -0.463 0.678 0.156
## srprs:AI-AI -0.335 -0.488 0.175 0.747 0.226
##
## Condition-specific novelty-to-resonance slopes:
## condition surprise.trend SE df lower.CL upper.CL
## Human-AI      1.068 0.0192 2535 1.030 1.106
## Human-Human    0.884 0.0367 2612 0.812 0.956
## AI-AI          0.891 0.0343 2600 0.824 0.959
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```



11 Supplemental: Role-Level View



12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from $\text{resonance} \sim \text{surprise} * \text{slot} + (1|\text{conversation_id})$. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story $A_{\text{influence}}$
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

- length-controlled re-fit (+ `nwords`) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in `analysis/human-ai/novelty.Rmd`)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-01 23:14:58.792535